

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
THEORY EXAMINATION

Question Paper

Month and year of the Exam August, 2026
Programme: B.Tech
Subject....Measurement and Instrumentation
Course No.... EEPEC212
Number of Questions to be Attempted....5
Total No. of Questions....6

Roll No. 124104072
Semester 4th
Maximum Marks 50
Time allowed 3 Hrs.
Total No. of Pages used::2

Unless stated otherwise, the symbols have their usual meanings in context with the subject. Assume suitably and state, additional data required, if any.

The candidates before starting to write the solutions, should check the question paper for any discrepancy, and also ensure that they have been delivered the question paper of right course no. and the right subject title.

Q.1.(a) Explain the following types of errors with suitable examples and remedies to reduce them: 5

- Instrumental errors
- Gross errors

Q.1.(b) Three resistors of 100Ω , 200Ω , and 300Ω , having limiting errors of $\pm 2\%$, $\pm 3\%$, and $\pm 4\%$, respectively, are connected in parallel. Determine 5

- The equivalent resistance
- Limiting error in percentage and in ohm.

Q.2.(a) Explain the ammeter-voltmeter method for measuring an unknown resistance. Draw neat circuit diagrams and derive the expressions for both configurations. Also, derive the conditions for selecting the appropriate configuration considering the internal resistances of the ammeter and voltmeter. 5

Q.2.(b) Define the sensitivity of a Wheatstone bridge. Derive the expression for bridge sensitivity and explain the factors affecting the sensitivity of a Wheatstone bridge. 5

Q.3.(a) Name and explain the AC bridge suitable for measuring a high-Q coil. Draw the circuit and phasor diagram, and derive the balance condition. Also obtain the expressions for the unknown inductance and resistance of the coil. 5

Q.3.(b) Derive the balance condition of a Schering bridge used to measure an unknown capacitance. Clearly draw the circuit and the phasor diagram. 5

The given parameters for the Schering bridge are:

Arm AB: A capacitor of $0.4\ \mu\text{F}$ connected in parallel with a $1.5\ \text{k}\Omega$ resistor.

Arm BC: A $3\ \text{k}\Omega$ resistor.

Arm CD: An unknown capacitor (C_x) in series with an unknown resistor (R_x).

Arm DA: A standard capacitor of $0.4\ \mu\text{F}$.

Supply frequency: $1\ \text{kHz}$.

At balance, determine:

- The value of the unknown capacitance (C_x).
- The value of the unknown resistance (R_x).
- The dissipation factor of the unknown capacitor.

Q.4.(a) I. Explain the formation of Lissajous patterns on a Cathode Ray Oscilloscope (CRO). Discuss the relationship between the frequencies and phase difference of 5

the two sinusoidal signals applied to the horizontal and vertical deflection plates.

- II. A sinusoidal signal of known frequency $f_1 = 2 \text{ kHz}$ is applied to the horizontal input of a CRO. An unknown-frequency sinusoidal signal is applied to the vertical input. The resulting Lissajous pattern shows 4 tangencies with the horizontal axis and 3 tangencies with the vertical axis. Determine the frequency of the unknown signal.

Q.4.(b) What is an instrument transformer? Explain its purpose and working principle. Explain the following parameters: 5

Transformation ratio, Turns ratio, Ratio error, Phase-angle error, Burden

Q.5.(a) I. Define the gauge factor and derive the relationship between gauge factor and poisson ratio. 5

- II. A strain gauge has an initial resistance of 120Ω and a gauge factor of 2.0. When subjected to a tensile strain of $800 \mu\epsilon$, determine:

- The change in resistance of the strain gauge.
- The final resistance of the strain gauge.
- The percentage change in resistance.

Q.5.(b) Explain the working principle, construction, and applications of the following transducers: 5

- Thermocouple
- Capacitive transducer

Also, mention their advantages and limitations.

Q.6.(a) Define Industry 4.0 and explain its significance in modern industrial automation. Discuss the major technologies associated with Industry 4.0. 5

Q.6.(b) Discuss the basic structure and working of IEEE 488 bus interfacing 5